

EDA VISUALIZATION

Building Queries and Visualization:

SQL:

```
USE CATALOG samples;  
SELECT  
    hour(tpcp_dropoff_datetime) as dropoff_hour,  
    COUNT(*) AS num  
FROM samples.nyctaxi.trips  
WHERE pickup_zip IN ('10001', '10002')  
GROUP BY 1;
```

Table ▼ +		
	¹ ₃ dropoff_hour	¹ ₃ num
1	15	71
2	20	88
3	21	80
4	22	60
5	17	54
6	1	40
7	23	60
8	4	20
9	13	69
10	8	42
11	6	19
12	3	18
13	16	63
14	11	53
15	0	57
↓ ▼ 24 rows 3.05s runtime		

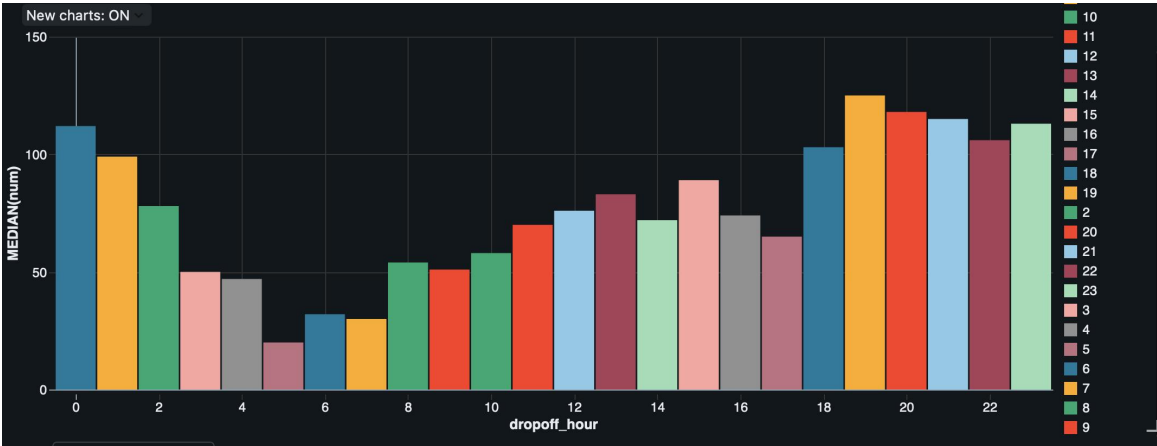
PYTHON :

```
from pyspark.sql.functions import hour, col

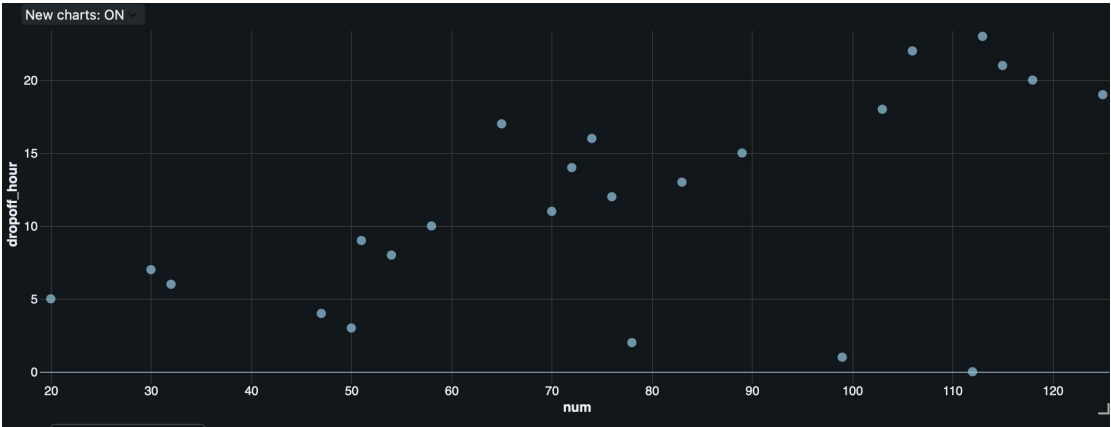
pickupzip = '10001' # Example value for pickupzip
df = spark.table("samples.nyctaxi.trips")
result_df = df.filter(col("pickup_zip") == pickupzip) \
               .groupBy(hour(col("tpep_dropoff_datetime")).alias("dropoff_hour")) \
               .count() \
               .withColumnRenamed("count", "num")
display(result_df)
```

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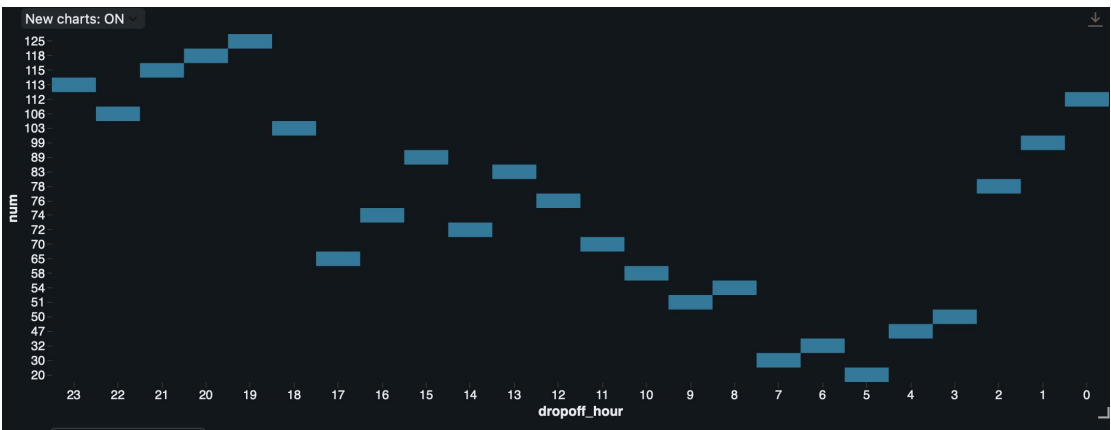
BAR CHART:



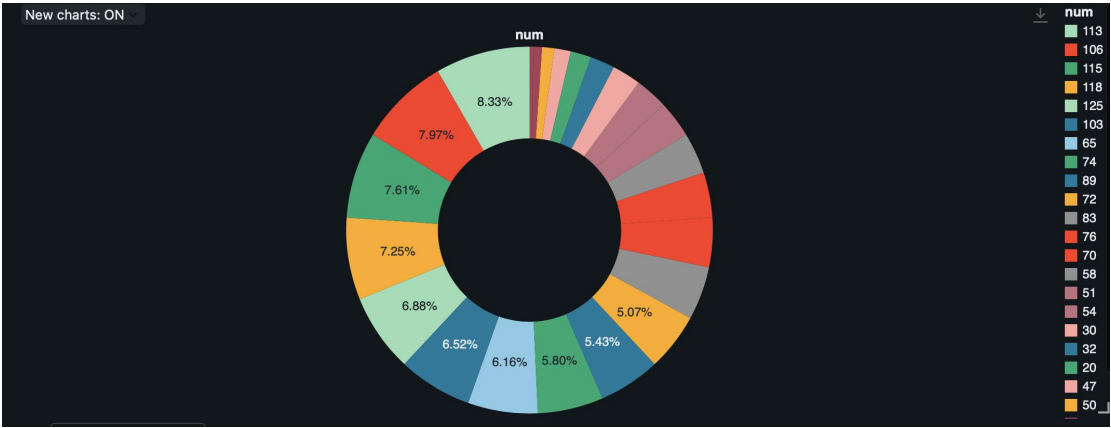
SCATTER PLOT:



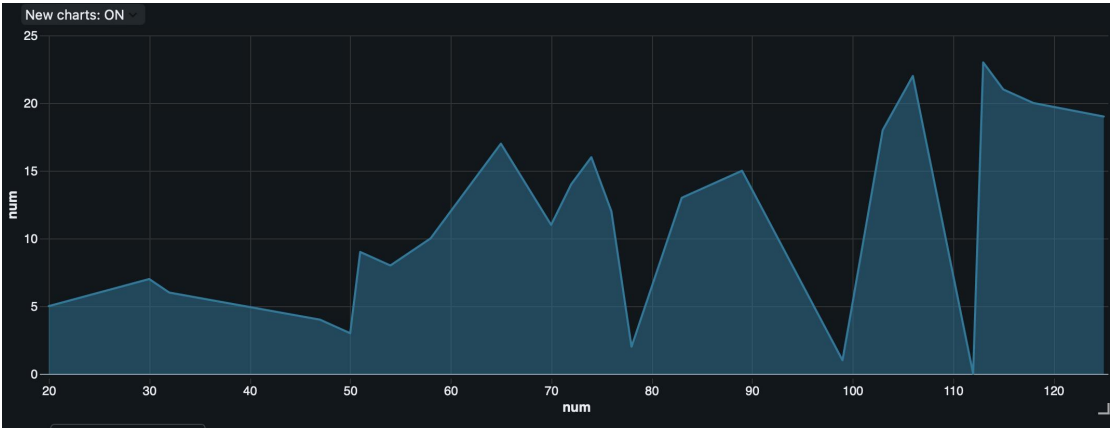
HEATMAP:



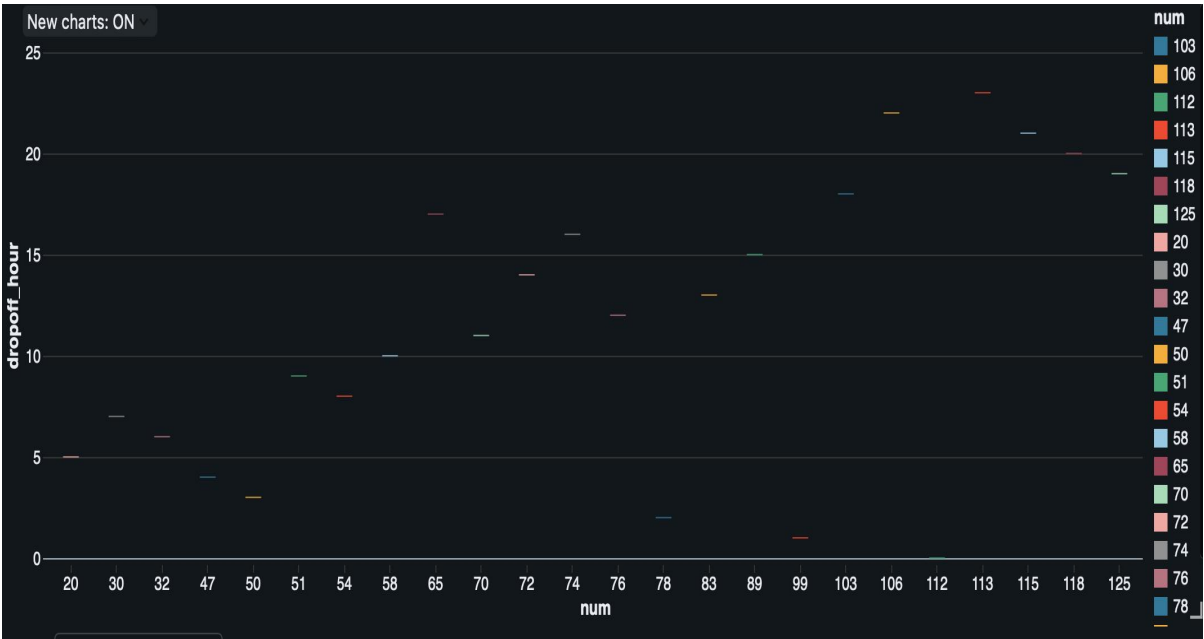
PIE CHART:



AREA PLOT:



BOX PLOT



COMBO PLOT:

